Forensic Science, Medicine and Pathology (2026) 22:645–651
https://doi.org/10.1007/s12024-025-01087-4

CASE REPORT

Forensic characteristics of starvation deaths: A case report and literature review

Chen Chen[1] · Zhonghao Yu[1] · Yuhao Yuan[1] · Xiangting Gao[1] · Qing Shi[1] · Yiwu Zhou[1]

Accepted: 29 August 2025 / Published online: 30 October 2025

Abstract

Starvation-related deaths are not uncommon worldwide particularly in economically underdeveloped regions. However, the forensic characterization of such cases remains limited due to underdeveloped local forensic systems. Here we presented a case of a neonate who died of starvation after being abandoned in the wild for 14 days, and 9 additional reported cases were identified by a systematic search of PubMed and MEDLINE databases. Based on these findings, the key forensic characteristics were summarized. Significant reductions in body or organ weight were consistently documented across all ten cases; PAS staining of the liver revealed markedly depleted glycogen stores in four cases. Of the four cases involving children under ten years of age, three demonstrated severe thymic atrophy. In our case, we observed pronounced calcification of thymic corpuscles. These findings suggest that thymic pathological changes may play a potential role in the forensic assessment of starvation deaths in children. In the absence of forensic standards for diagnosing starvation deaths, our case report and consolidated review provide valuable reference points for the forensic evaluation of similar incidents.

Keywords Forensic pathology · Starvation · Systematic review · Organ weight reduction · Thymic calcification

Introduction

Starvation remains a prevalent issue globally. As of 2023, it is estimated that 9.1% of the world's population suffers from hunger, meaning that approximately one in eleven people are at risk of death due to starvation [1]. A recent study investigating causes of death in Ethiopia's northern Tigray region identified starvation as a leading cause of mortality across all age groups, with fatalities predominantly occurring among children under five, primarily due to severe malnutrition [2]. Despite the numerous reported fatalities, research on the associated forensic features remains extremely rare, likely reflecting underdeveloped forensic systems in economically disadvantaged regions. Given the rarity of such case reports and guidelines, we presented the case of a neonate who survived 14 days of starvation before succumbing to severe malnutrition. Through a systematic search of the PubMed and MEDLINE databases, we identified and analyzed nine similar reports, synthesizing the histopathological changes observed across multiple organs.

Chen Chen and Zhonghao Yu are contributed equally to this work.

✉ Chen Chen
hust_chenc@163.com

✉ Zhonghao Yu
1259970467@qq.com

✉ Yiwu Zhou
zhouyiwu@hust.edu.cn

Yuhao Yuan
362736952@qq.com

Xiangting Gao
1404467629@qq.com

Qing Shi
806496165@qq.com

1 Department of Forensic Medicine, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, China

Case report

Case history

In June 2024, a male neonate with weak vital signs was discovered in a secluded wooded area. Despite emergency intervention, the infant was pronounced dead. Subsequent investigations revealed that the neonate had been born 14 days prior and abandoned by relatives on the second day after birth due to bilateral cleft lip. Environmental data recorded an average temperature of 26 .02 ± 5.71 °C and a relative humidity of 63.94 ± 11.65% over the preceding 15 days, while on-site examinations confirmed that the location was notably remote.

Clinical records

The term neonate, born at 40 weeks of gestation, had a birth weight of 3510 g, with a body temperature of 36.3 °C. The infant exhibited the typical appearance of a full-term newborn, was well-nourished, and demonstrated stable respiration. Sucking, rooting, Moro, and grasp reflexes were all intact. Blood glucose levels remained stable, with normal urination and defecation. Additionally, the neonate presented with bilateral complete cleft lip.

Autopsy and histopathology findings

External examination The body of a male neonate weighing 2490 g was examined. The subcutaneous fat layer was notably thin, measuring only 1 mm. Bilateral cleft lip was present, with dimensions of 2.0 cm × 1.5 cm. Scabbed abrasions and contusions were observed on the left shoulder and left thigh. No other significant external abnormalities were identified.

Internal examination and histopathology The brain weighed 418 g and showed severe cerebral edema. Numerous neurons demonstrated nuclear pyknosis, loss of Nissl substance, and increased cytoplasmic eosinophilia, indicative of ischemic-hypoxic injury (Fig. 1a). In addition, several small blood vessels contained hyaline thrombus. The thymus, weighing 2.0 g, was markedly atrophic. Histologically, multiple foci of thymic corpuscle calcification were observed, accompanied by significant vascular congestion within the interstitium (Fig. 1b). The heart weighed 26.0 g microscopic evaluation revealed focal areas of myocardial nuclear pyknosis and necrosis along with interstitial vascular congestion.The lungs weighed 21.0 g (left) and 23.0 g (right). Microscopically, multifocal atelectasis was apparent, some alveoli contained keratinized epithelium, meconium, and red blood cells. Additionally, there was congestion in the capillaries of the alveolar walls and small interstitial vessels, with hyaline thrombus present in certain vessels (Fig. 1c). The kidneys, with a combined weight of 22.0 g, displayed extensive proteinaceous and granular casts within the renal tubules and collecting ducts, along with focal necrosis and detachment of some tubular epithelial cells (Fig. 1d). The liver, weighing 79.0 g, had a dark green cut surface. Histological analysis revealed hepatocyte edema, with a subset of hepatocytes showing increased cytoplasmic eosinophilia and nuclear pyknosis (Fig. 1e). Bile stasis was observed within the hepatic sinusoids, and Periodic acid-Schiff (PAS) staining for hepatic glycogen was negative (Fig. 1f). The spleen, weighing 4.0 g, exhibited marked sinusoidal congestion.

Toxicological analysis

Routine toxicological analysis for ethanol, rodenticides, organophosphate pesticides, and other common toxicants was performed on the liver, stomach, and gastric contents, with all tests yielding negative results.

Forensic diagnosis

Based on a comprehensive evaluation of the case details, histopathological findings, and toxicological results, it was concluded that the neonate suffered from severe malnutrition due to insufficient nutritional intake, ultimately resulting in multi-organ failure and death.

Literature search strategy

A systematic review of the literature was performed from 2000 in accordance to the PRISMA statement [3]. Database search included PubMed and MEDLINE. Keywords and combinations were related to “starvation death” “starvation and autopsy”.

Duplicate records were removed manually using Mendeley. Irrelevant titles or non-English articles were excluded. We then reviewed abstracts if available and full-text articles; articles were excluded if they were irrelevant, not case reports, or did not identify starvation or malnutrition as the cause of death. Studies involving animals, as well as editorials, technical notes, commentaries and opinion pieces were also excluded (Fig. 2).

Results

After excluding unrelated studies, nine studies were identified through our review. Ten individuals who died of starvation in this study were summarized in Table 1. The majority were elderly or children, with ages ranging from 14 days to 95 years. 60% had no past medical history and neglect was the most common cause of death (70%). Half of the reports specified the duration of starvation, which ranged from 2 to 70 days. In every pediatric case, recorded body weight was at least 30% below the normal weight for age, with an average reduction of 38%.

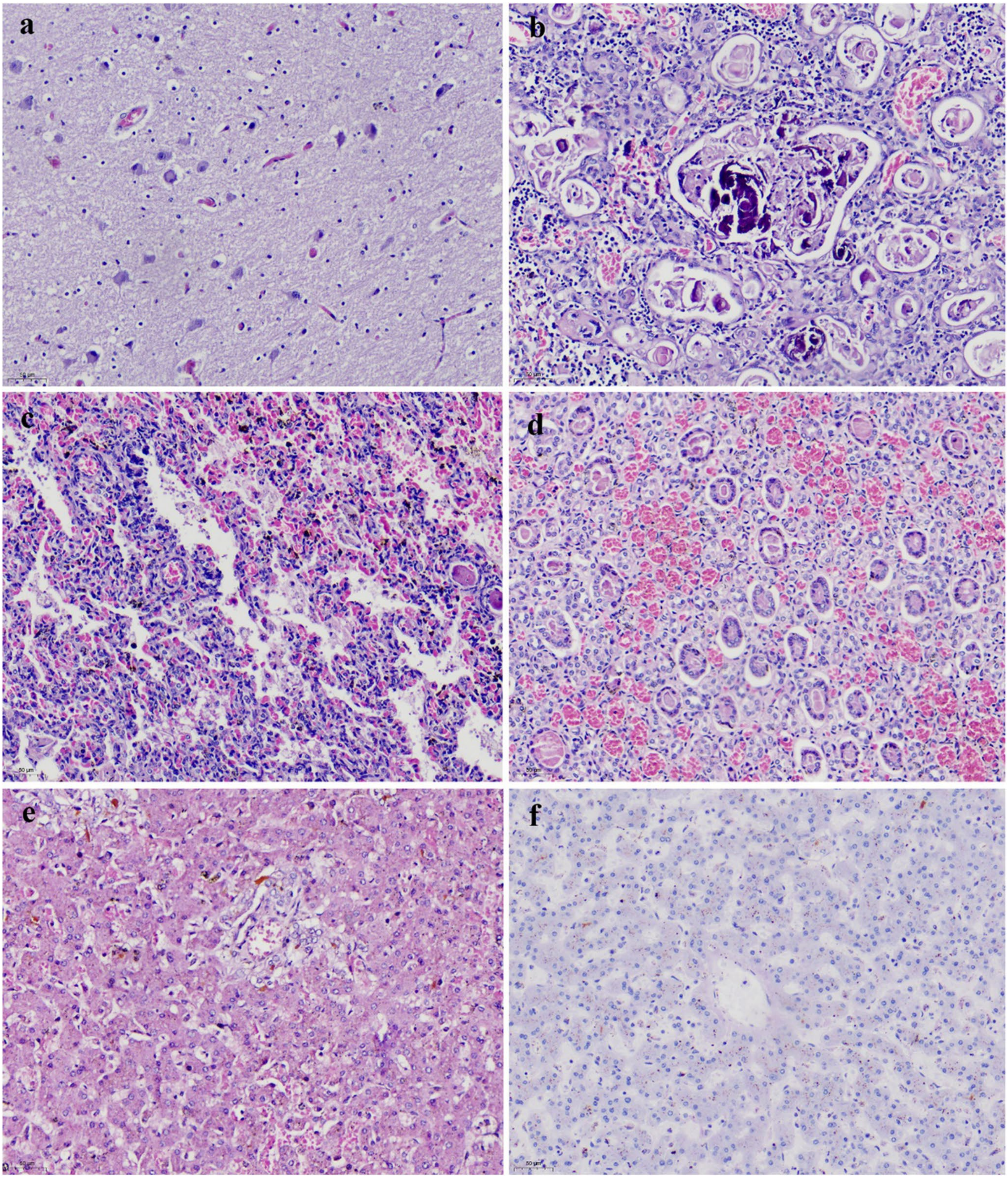

Fig. 1 (**a**) Ischemic-hypoxic changes in cerebral neurons (H&E); (**b**) Calcification of thymic corpuscle (H&E); (**c**) Pulmonary atelectasis with hyaline thrombus (H&E); (**d**) Proteinaceous casts in renal tubules (H&E); (**e**) Hepatocellular necrosis with bile stasis (H&E); (**f**) Depletion of hepatic glycogen stores (PAS)

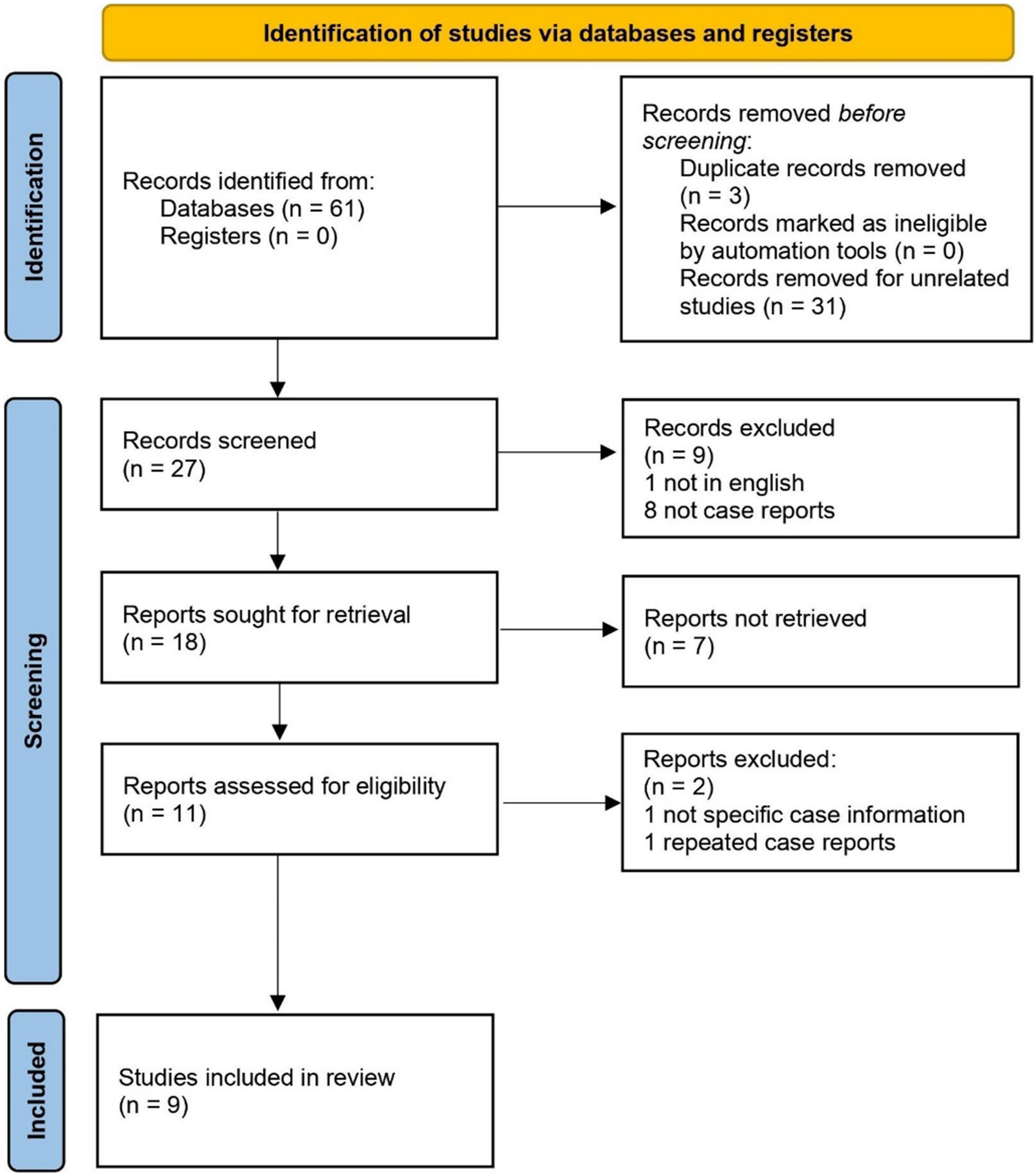

Fig. 2 Flowchart of searched, selected and excluded articles following the PRISMA statement

Table 1 Review of the literature: reported cases of starvation to death

Case ID	Age	Sex	Time of starvation	Cause of starvation	Weight of the corpse(kg)	Weight loss (%)	Gross features	Microscopic features
1 4	3y	F	2d	Neglect	9	36%	Reduced organ weight	Thymic atrophy, steatosis of the liver, myocyte vacuolation and Armanni-Ebstein lesion
2 5	16 m	NA	NA	Neglect	6.08	40%	Extreme dehydration, sunken eyes, dry and wrinkly skin	Fatty hepatic degeneration, severe glycogen depletion (Periodic acid-Schiff)
3 6	95y	M	NA	Neglect	NA	NA	Reduced organ weight, thin and dry skin, eyeballs were sunken, buccal fat pads were completely lost, and oral mucosa was dry	Cerebral edema, severe glycogen depletion (Periodic acid-Schiff)
47	76y	F	NA	Neglect	27.2	NA	Severe weight loss	NA
58	16 m	F	1w	Abuse	5.7	42%	Reduced organ weight, severe dehydration, obvious bony prominence and sunken eyes	Thymic atrophy, massive oedema of the brain, hepatic steatosis and no glycogen in the liver (Periodic acid-Schiff)
6 9	10y	M	1w	Neglect	16.8	NA	Reduced organ weight skin turgor was diminished and the thickness of subcutaneous adipose tissue was 0 .8 cm	Severe hepatic steatosis
7 10	47y	F	NA	Anorexia nervosa	24	NA	Severely cachectic, obvious muscle atrophy and reduced organ weight	Increased lipofuscin in cardiomyocytes, diffuse endocardial and interstitial fibrosis
811	26y	M	70d	Hunger strike	39	NA	Obvious muscle wasting with reduced subcutaneous fat (severe cachexia)	Multiple organ atrophy
9 12	47y	F	NA	Neglect	35	NA	A marked reduction of both subcutaneous and visceral adipose tissue, and obvious muscle atrophy	Significant cellular shrinkage, and mucoid interstitial alteration was found in the adipose tissue
10	14d	M	14d	Neglect	2.49	34%	The subcutaneous fat layer was notably thin, measuring 1 mm, and reduced organ weight	Thymic atrophy, calcification of thymic corpuscle, and depletion of hepatic glycogen stores (Periodic acid-Schiff)

M male, F female, d day, w week, m month, y year, NA Not applicable

Autopsies were performed in all cases; only one underwent an external examination without histopathology. Every report detailed reductions in body or organ weight and the majority (70%) noted loss of adipose tissue and severe dehydration. Histopathological evaluation using PAS staining was recorded in 40% of cases, all of which demonstrated marked glycogen depletion. Of the four cases involving children under ten years of age, three reported thymic atrophy, affecting individuals aged 3 years, 16 months, and 14 days (Table 1).

Discussion

In our era, which places a high premium on human rights, it is disheartening that such fatal cases of starvation, whether due to neglect or other causes, are not uncommon, even in developed countries [13, 14]. Despite its global prevalence detailed case reports and literature on fatal outcomes remain exceedingly scarce and deaths resulting from neglect continue to be overlooked. Consequently investigating starvation deaths has become a significant challenge for forensic pathologist due to the paucity of available data. The report presented a reference case involving a neonate who endured 14 days of starvation before ultimately succumbing to it. Existing literature has documented the remarkable resilience of neonates under extreme conditions of food deprivation, although such efforts are often ultimately futile [15]. Furthermore, a systematic search of the PubMed and MEDLINE databases was conducted, identifying nine additional reported cases of starvation-related deaths, which were collectively analyzed to summarize their forensic characteristics.

In forensic autopsies of neonates, the preliminary collection of data, including clinical records [16], geographical location [17]and environmental temperature [18]is crucial for excluding concurrent causes of death or underlying conditions (e.g. malabsorption disorders). In this case, the neonate was found abandoned in an extremely remote area, and the presence of bilateral cleft lip rendered feeding impossible without medical intervention. Over the preceding 15 days, the local average temperature was 26 .02±5.71 °C, with a relative humidity of 63.94±11.65%, such conditions are insufficient to induce death from either heat or cold stress in a neonate [19–21].

Autopsy findings revealed a marked decrease in body weight from 3510 g to 2490 g, a reduction of 1020 g (29.1%), with a subcutaneous fat thickness of only 1 mm. Compared to average organ weights of neonates with similar birth weight, the lungs, liver, spleen, kidneys, and thymus all exhibited varying degrees of weight loss, with thymic degeneration being the most pronounced (an 84.1% reduction), which is one of the most characteristic features of neonatal starvation [22, 23]. Correspondingly, histopathological examination revealed calcification of thymic corpuscles, possibly due to the heightened sensitivity of immature double-positive (DP) thymocytes to chronic stressors such as prolonged starvation, leading to a selective reduction in thymic cell numbers [15, 24]. Since the thymus undergoes natural involution after the first year of life and the population of DP cell declines significantly with age [25, 26], explaining thymic corpuscle calcification is rarely mentioned in cases of starvation deaths in other age groups [12, 27]. Further analysis using PAS staining on the liver demonstrated a markedly negative result for hepatic glycogen, indicating depleted glycogen stores and energy exhaustion [27]. In the kidneys, H&E staining revealed extensive formation of proteinaceous and granular casts within the renal tubules and collecting ducts, suggesting severe renal damage, which was consistent with previous findings [11, 28]. Furthermore, evidence of multi-organ failure was observed, including ischemic-hypoxic changes in cerebral neurons and hepatocellular necrosis. Pulmonary hyaline thrombus was also noted, which may indicate terminal microcirculatory and coagulation disturbances in the context of severe malnutrition and dehydration [29, 30]. The limitation of the case is that we are unable to reconstruct the specific process of starvation, including whether dew or rainwater was ingested. Nevertheless, based on the background investigation of the case and the pathological changes observed during the autopsy, we have concluded through comprehensive analysis that the newborn was in a state of prolonged starvation. Ultimately, the cause of death was determined to be multi-organ failure resulting from severe malnutrition.

Table 1summarizes reported cases of death due to starvation since 2000. Given the limited number of cases, we included cases of starvation deaths among both adults and the elderly. Apart from the present case, none involved neonatal fatalities. It is noteworthy that most of the case reports lacked detailed pathological descriptions for reference. Among the cases where death from starvation was suspected, only 40 % documented the use of Periodic acid-Schiff (PAS) staining. These findings highlight that, unlike other causes of death, cases of starvation still lack systematic guidelines and standardized reference protocols.

Our case exemplified neonatal death resulting from starvation. In a review of ten reported cases, including our own, we observed that weight loss, either in the body as a whole or in individual organs, was a prominent and consistent feature across all age groups. Additionally, severe loss of adipose tissue and dehydration were frequently reported and may be useful indicators of the severity of starvation. In all cases where Periodic acid-Schiff (PAS) staining was performed, a significant reduction in hepatic glycogen stores was observed. This suggests that glycogen depletion is one of the characteristic pathological changes in starvation-related deaths and that PAS staining provides important diagnostic information in these cases. In pediatric starvation cases, severe pathological atrophy of the thymus appears to be a distinct and age-specific finding. In our case, calcification of thymic corpuscles was observed, indicating that thymic pathology may play a potential role in the forensic evaluation of starvation deaths in children. In forensic practice, the postmortem and histopathological assessment of starvation-related deaths often lacks standardized procedures and criteria for interpretation. Based on the review of ten reported cases, we propose several recommendations to support forensic evaluations in similar circumstances. Firstly, meticulous attention should be paid to variations in body weight and individual organ weights during autopsy, with particular emphasis on thymic alterations in pediatric cases. Secondly, histopathological analyses should include hepatic fat and glycogen staining, particularly PAS staining, in all suspected starvation deaths. Lastly, a comprehensive exclusion of traumatic injuries is imperative. Integrating investigative findings with medical records facilitates an accurate determination of the cause of death.

Conclusion

We reported a case of a neonate who died of starvation after being abandoned for 14 days and we systematically reviewed and summarized the forensic characteristics of 10 reported cases of starvation-related deaths. Among them, four cases showed a marked reduction in hepatic glycogen content as demonstrated by PAS staining. Of the four cases involving children under the age of 10, three reported severe pathological thymic atrophy. In our case, prominent calcification of the thymic corpuscles was also observed, suggesting that thymic pathology may play a potential role in the forensic evaluation of starvation deaths in children. Therefore, we would like to present this paper to forensic workers in the hope of providing some reference value for forensic medical identification of similar cases.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s12024-025-01087-4.

Funding There was no funding for this work.

Declarations

Ethical Approval The abandoned infant was found in the wild. We conducted an autopsy on the deceased infant at the request of the local law enforcement authorities, which was approved by the local administrative and judicial authorities. The autopsy was conducted in strict accordance with Forensic medicine—Specifications for examination of neonate cadavers (China, GA/T 151–2019) to ensure compliance with ethical requirements. After the autopsy, the body was cremated by the law enforcement authorities, except for the necessary tissue samples.

Conflict of interest The authors declare they have no conflict of interests.

References

1. The State of Food Security and Nutrition in the World 2024. FAO; IFAD; UNICEF; WFP; WHO. 2024. https://doi.org/10.4060/cd1254en
2. Haileselassie M, Kahsay H, Teklemariam T, et al. Starvation remains the leading cause of death in Tigray, Northern Ethiopia, after the Pretoria deal: a call for expedited action. BMC Public Health. 2024;24(1):3413. https://doi.org/10.1186/s12889-024-20932-9.
3. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews | the BMJ. Accessed April 24. 2025. https://www.bmj.com/content/372/bmj.n71.long
4. Milroy CM, Parai JL. Armanni-Ebstein lesion, ketoacidosis and starvation in a child. Forensic Sci Med Pathol. 2011;7(2):213–6. https://doi.org/10.1007/s12024-010-9213-6.
5. Palmiere C, Staub C, La Harpe R, Mangin P. Parental substance abuse and accidental death in children. J Forensic Sci. 2010;55(3):819–21. https://doi.org/10.1111/j.1556-4029.2010.01349.x.
6. Baković M, Vreča P, Mayer D. Case of fatal starvation: can stable isotope analysis serve to support morphological diagnosis and approximate the length of starvation? J Forensic Sci. 2017;62(1):258–64. https://doi.org/10.1111/1556-4029.13244.
7. Akaza K, Bunai Y, Tsujinaka M, et al. Elder abuse and neglect: social problems revealed from 15 autopsy cases. Leg Med. 2003;5(1):7–14. https://doi.org/10.1016/S1344-6223(02)00057-3.
8. Solarino B, Grattagliano I, Catanesi R, Tsokos M. Child starvation and neglect: a report of two fatal cases. J Forensic Leg Med. 2012;19(3):171–4. https://doi.org/10.1016/j.jflm.2011.10.004.
9. Yamada A, Unuma K, Arai N, Kitamura O, Uemura K. Inappropriate diet and fatal malnutrition in a 10-year-old child fed only infant formula throughout life: novel pathological diagnostic criterion for starvation via lipophagy. Forensic Sci Int. 2021;325:110896. https://doi.org/10.1016/j.forsciint.2021.110896.
10. Lamzabi I, Syed S, Reddy VB, Jain R, Harbhajanka A, Arunkumar P. Myocardial changes in a patient with anorexia nervosa: A case report and review of literature. Am J Clin Pathol. 2015;143(5):734–7. https://doi.org/10.1309/AJCP4PLFF1TTKENT.
11. Altun G, Akansu B, Ugur Altun B, Azmak D, Yilmaz A. Deaths due to hunger strike: Post-mortem findings. Forensic Sci Int. 2004;146(1):35–8. https://doi.org/10.1016/j.forsciint.2004.03.022.
12. Franchetti G, Cestonaro C, Giordano R, et al. Severe starvation and restraint in a 47-year-old woman: clinical, autopsy and histopathological evidence of abuse and neglect. Forensic Sci Int. 2024;355:111941. https://doi.org/10.1016/j.forsciint.2024.111941.
13. Brasher MI, Silverstein A, Moher J. Evaluation and management of an abandoned infant at a quaternary care hospital. J Emerg Med. 2022;62(3):e69–76. https://doi.org/10.1016/j.jemermed.2021.11.006.
14. Makhlouf F, Rambaud C. Child homicide and neglect in France: 1991–2008. Child Abuse Negl. 2014;38(1):37–41. https://doi.org/10.1016/j.chiabu.2013.08.016.
15. Madea B, Ortmann J, Doberentz E. Forensic aspects of starvation. Forensic Sci Med Pathol. 2016;12(3):276–98. https://doi.org/10.1007/s12024-016-9777-x.
16. Hofer P, Brandau LM, Mützel E. Rechtsmedizinische aspekte der kindesmisshandlung. Radiologe. 2016;56(5):432–7. https://doi.org/10.1007/s00117-016-0101-2.
17. Gheorghe A, Banner J, Hansen SH, Stolborg U, Lynnerup N. Abandonment of newborn infants: a Danish forensic medical survey 1997–2008. Forensic Sci Med Pathol. 2011;7(4):317–21. https://doi.org/10.1007/s12024-011-9253-6.
18. Karlsson L, Lundevaller E, Schumann B. The association between cold extremes and neonatal mortality in Swedish Sápmi from 1800 to 1895. Glob Health Action. 2019;12(1):1623609. https://doi.org/10.1080/16549716.2019.1623609.
19. Basagaña X, Sartini C, Barrera-Gómez J, et al. Heat waves and cause-specific mortality at all ages. Epidemiology. 2011;22(6):765. https://doi.org/10.1097/EDE.0b013e31823031c5.
20. Karlsson L, Lundevaller EH, Schumann B. Neonatal mortality and temperature in two Northern Swedish rural parishes, 1860–1899—the significance of ethnicity and gender. Int J Environ Res Public Health. 2020;17(4):1216. https://doi.org/10.3390/ijerph17041216.
21. Lakhoo DP, Blake HA, Chersich MF, Nakstad B, Kovats S. The effect of high and low ambient temperature on infant health: A systematic review. Int J Environ Res Public Health. 2022;19(15):9109. https://doi.org/10.3390/ijerph19159109.
22. Guleria P, Jain D. Thymic lesions of the paediatric age group: A comprehensive review of non-neoplastic and neoplastic etiologies. Mediastinum. 2019;3(0). https://doi.org/10.21037/med.2019.05.04.
23. Flomenbaum MA, Warner RC. Morphologic markers of acute and chronic stress in child abuse. Am J Clin Pathol. 2022;157(6):823–35. https://doi.org/10.1093/ajcp/aqab204.
24. Dinges SS, Amini K, Notarangelo LD, Delmonte OM. Primary and secondary defects of the thymus. Immunol Rev. 2024;322(1):178–211. https://doi.org/10.1111/imr.13306.
25. Thapa P, Farber DL. The role of the thymus in the immune response. Thorac Surg Clin. 2019;29(2):123–31. https://doi.org/10.1016/j.thorsurg.2018.12.001.
26. Liang Z, Dong X, Zhang Z, Zhang Q, Zhao Y. Age-related thymic involution: mechanisms and functional impact. Aging Cell. 2022;21(8):e13671. https://doi.org/10.1111/acel.13671.
27. Garland J, Irvine R. A guide to the postmortem investigation of starvation in adults. Am J Forensic Med Pathol. 2022;43(3):205–14. https://doi.org/10.1097/PAF.0000000000000763.
28. Navis G. Consequences of famine and malnourishment on kidney health. Nat Rev Nephrol. 2023;19(3):139–40. https://doi.org/10.1038/s41581-023-00681-5.
29. Preidis GA, Soni KG, Suh JH, et al. Coagulopathy in malnourished mice is sexually dimorphic and regulated by nutrient-sensing nuclear receptors. Hepatol Commun. 2020;4(12):1835–50. https://doi.org/10.1002/hep4.1622.
30. Kimhi G, Rubinshtein M, Tirosh-Wagner T, Mishaly D, Kenet G, Paret G. Dehydration as a rare cause of pulmonary artery thrombosis in a 2-week-old term neonate. J Pediatr Intensive Care. 2018;7(2):102–5. https://doi.org/10.1055/s-0037-1607445.